

RoomReady: Study Room Reservation System - SRS Document

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MCS 5308: Software Engineering

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for

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SOFTWARE REQUIREMENTS SPECIFICATION

Purpose

This document is intended for the RoomReady engineering team (Aaron, Aaron, and Valerie) and project faculty. It defines the functional and non-functional requirements for the RoomReady Proof of Concept (POC), supporting architectural decisions, development prioritization, and project scope management for the 5-hour Saturday build and Sunday presentation.

Scope

RoomReady is a kiosk-style web application designed for public library patrons to identify and reserve currently available study rooms.

- **In-Scope:** Current room availability display, room selection, reservation form submission, confirmation messaging, and immediate status updates from Available to Reserved using simulated frontend data.
- **Out-of-Scope:** User authentication, long-term calendar management, recurring reservations, physical room access control, payment processing, production database integration, and full library event scheduling.

Definitions and Abbreviations

Term/Abbreviation	Definition
FR/NFR	Functional Requirement/Non-Functional Requirement
SRS	Software Requirements Specifications
RTM	Requirements Traceability Matrix
Patron	A public library user who wants to reserve a study room.
Availability Dashboard	The main screen that displays each study room and its current Available or Reserved status.
Simulated Data	Mock room data stored in the frontend prototype to demonstrate reservation behavior without a production backend.

SOFTWARE REQUIREMENTS SPECIFICATION

Table 1. *Defines the technical terminology and acronyms used throughout this Software Requirements Specification*

References

- Graf, A., Haywood, V. A., & Tobias, A. (2026, June 13). *RoomReady: Study Room Reservation System - Vision Document v1.0*. Concordia University Texas.
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Overall Description

Product Overview

RoomReady is a user-friendly kiosk-style web application that helps public library patrons view current study room availability and reserve an open room through a simple walk-up process. For the proof of concept, the system uses a frontend-only prototype with simulated room data to demonstrate the core reservation workflow. When a patron submits a reservation, the selected room status updates from Available to Reserved in the interface, showing how the system would reduce staff interruptions and make room availability easier to understand.

Stakeholder Analysis

Stakeholder	Role	Interest in the System
Library Patrons	Easily reserve a study room without staff mediation.	Limited by the 1-hour/day policy.
Library Staff	Reduce administrative time spent on manual scheduling.	Must remain within the 4-person room capacity limit.
Engineering Team	Build a functional POC within a 5-hour Saturday sprint.	Limited to free-tier tools and defined tech stack.
Library Event Coordinator	Prevent study room bookings from overlapping with scheduled library workshops or events.	Need advanced visibility into room availability for planning purposes.

Table 2. Identifies the key individuals and groups influenced by RoomReady, along with their specific interests in the system's functionality.

SOFTWARE REQUIREMENTS SPECIFICATION

Functional Requirements

Room Availability & Reservation Flow

ID	Requirement	Priority	Related Story
FR-01	The system shall display a dashboard listing public library study rooms.	Must Have	US-001
FR-02	The system shall show each study room with a current status of Available or Reserved.	Must Have	US-001
FR-003	The system shall allow patrons to select a room marked Available.	Must Have	US-002
FR-004	The system shall prevent patrons from selecting a room marked Reserved.	Must Have	US-002
FR-005	The system shall display a reservation form for the selected available room.	Must Have	US-003
FR-006	The system shall require the patron to enter basic reservation details before submitting the form.	Must Have	US-003
FR-007	The system shall display a confirmation message after successful submission.	Must Have	US-004
FR-008	The system shall update the selected room status from Available to Reserved after submission.	Must Have	US-004
FR-009	The system shall route	Should Have	US-004

SOFTWARE REQUIREMENTS SPECIFICATION

	the patron back to the dashboard after the reservation is completed.		
FR-010	The system shall use simulated frontend data to demonstrate room status changes without a production database.	Must Have	US-005

Table 3. Lists the functional requirements for the RoomReady proof of concept.

Non-Functional Requirements

Performance

ID	Category	Requirement	Measurement
NFR-01	Performance	The dashboard shall display room availability and update room status without noticeable delay during the demo	Status change appears immediately after form submission and returns to the dashboard.
NFR-02	Security/Privacy	The prototype shall avoid collecting sensitive personal information beyond basic reservation details.	No password, payment information, or library account authentication is required in the POC.
NFR-03	Usability	A first-time patron shall be able to complete the reservation flow with minimal instruction.	Reservation flow can be completed in under 3 minutes during the live demo.
NFR-04	Reliability	The prototype shall remain usable in a modern web browser for the Sunday presentation.	The app launches successfully from the /poc folder before the live demo.

Table 4. Defines the measurable quality attributes of RoomReady for a public library proof-of-concept environment.

Requirements Traceability Matrix (RTM)

ID	Requirement Summary	User Story IDs	Status
FR-001	Display study room dashboard	US-001	Planned
FR-002	Show Available/Reserved status	US-001	Planned
FR-003	Select an available room	US-002	Planned
FR-004	Prevent selection of reserved rooms	US-002	Planned
FR-005	Display reservation form	US-003	Planned
FR-006	Submit basic reservation details	US-003	Planned
FR-007	Show confirmation message	US-004	Planned
FR-008	Update room status to Reserved	US-004	Planned
FR-009	Route patron back to dashboard	US-005	Planned
FR-010	Use simulated frontend data	US-005	Planned

Table 5. Maps functional requirements to related user stories to show alignment between the product backlog and the RoomReady proof of concept.

Ethical Use of AI

As a novice in the field of computer science with no prior professional industry experience, we used Gemini (AI) and Grammarly to support the development of this SRS Document and related project artifacts.

- *Technical Guidance & Troubleshooting:* We relied on Gemini to navigate environment setup and troubleshoot configurations, which helped bridge the gap between academic learning and practical application of these tools for the RoomReady project.

SOFTWARE REQUIREMENTS SPECIFICATION

- *Structural & Technical Writing:* Gemini helped outline the report and draft complex sections such as the Success Criteria, ensuring it met professional software engineering standards (Sommerville, 2016).
- *Documentation Standards:* We used Gemini to ensure all headings, table captions, and citations follow the APA 7th Edition format.
- *Editing & Clarity:* Grammarly was used to review the document for grammatical accuracy, tone consistency, and overall readability, ensuring the final deliverable meets graduate-level communication standards.

Graf, Haywood, and Tobias made all final decisions on the system's architecture, features, and governance to ensure the RoomReady system aligns with the goals of the MCS 5308 course.